Scalability & Future Plan

Date	21 October 2026
Team ID	EN25CS3010849
Project Name	Global Olympic Sports Performance Analysis
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	**Static Dataset Dependency:** Relies on downloaded historical CSV files (medals.csv, athlete_events.csv) which do not update automatically.	New Olympic games data must be manually appended and cleaned.	High
2	**Single-User BI Environment:** Dashboards are designed for standalone analyst desktop review rather than multi-user cloud collaboration.	Team members or external stakeholders cannot view live interactive reports simultaneously over the web without local file access.	Medium
3	**Basic Historical Scope:** Focuses primarily on past Olympic editions without predictive analytics or future medal forecasting models.	Limited capability to predict upcoming athlete performance or future medal standings.	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Standalone desktop Tableau / Power BI viewer.	Deploy dashboards to Tableau Cloud / Power BI Service to support multi-user concurrent access.
2	Data Storage	Local static CSV files (~270k records).	Migrate processed data into a cloud relational database (PostgreSQL / MySQL) for optimized querying.
3	Performance	Local Pandas dataframe processing and filter rendering (~0.5s response).	Implement database indexing and pre-aggregated summary tables to handle massive future datasets instantly.
4	Security	Local file access security.	Introduce role-based access control (RBAC) and secure authentication via enterprise cloud gateways.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact